

Tree & Graph DFS/DP LeetCode Roadmap (CP Focus)

Level 1: Basic Tree DFS

- LC 559 - Maximum Depth of N-ary Tree
- LC 1522 - Diameter of N-ary Tree
- LC 1245 - Tree Diameter
- LC 690 - Employee Importance
- LC 1443 - Minimum Time to Collect All Apples in a Tree

Level 2: Subtree DP

- LC 1519 - Number of Nodes in the Subtree With the Same Label
- LC 2049 - Count Nodes With the Highest Score
- LC 2003 - Smallest Missing Genetic Value in Each Subtree
- LC 1948 - Delete Duplicate Folders in System
- LC 3331 - Find Subtree Sizes After Changes

Level 3: Tree DP

- LC 2246 - Longest Path With Different Adjacent Characters
- LC 834 - Sum of Distances in Tree
- LC 2581 - Count Number of Possible Root Nodes
- LC 2538 - Difference Between Maximum and Minimum Price Sum
- LC 2646 - Minimize the Total Price of Trips

Level 4: Hard Tree DP

- LC 1617 - Count Subtrees With Max Distance Between Cities
- LC 310 - Minimum Height Trees
- LC 2603 - Collect Coins in a Tree
- LC 2467 - Most Profitable Path in a Tree
- LC 2440 - Create Components With Same Value

Level 5: Graph DFS

- LC 785 - Is Graph Bipartite
- LC 802 - Find Eventual Safe States
- LC 841 - Keys and Rooms
- LC 797 - All Paths From Source to Target
- LC 1557 - Minimum Number of Vertices to Reach All Nodes

Level 6: Graph DP

- LC 1857 – Largest Color Value in a Directed Graph
- LC 2328 – Number of Increasing Paths in a Grid
- LC 329 – Longest Increasing Path in a Matrix
- LC 2065 – Maximum Path Quality of a Graph
- LC 1786 – Number of Restricted Paths

Advanced CP Topics

1. Tree Height & Diameter
2. Subtree DP
3. Longest Path DP
4. Rerooting DP
5. Euler Tour
6. Binary Lifting (LCA)
7. DSU on Tree
8. Heavy-Light Decomposition (HLD)
9. Centroid Decomposition

Recommended 15 Problems

1. LC 559
2. LC 690
3. LC 1245
4. LC 1522
5. LC 1443
6. LC 1519
7. LC 2049
8. LC 2246
9. LC 834
10. LC 2581
11. LC 2538
12. LC 2646
13. LC 2467
14. LC 2440
15. LC 1617